

Electricity Prices forecasting based on Energy Consumption and Weather Conditions

DS5110: Final Project Team 9
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Introduction

Background

Rising concerns over global warming drive the need for more efficient and sustainable electricity distribution. How is Spain's electricity market is influenced by energy consumption, generation, and weather conditions.

Problem/Motivation

Forecasting electricity prices accurately is vital for better energy management and cost reduction.

Goal

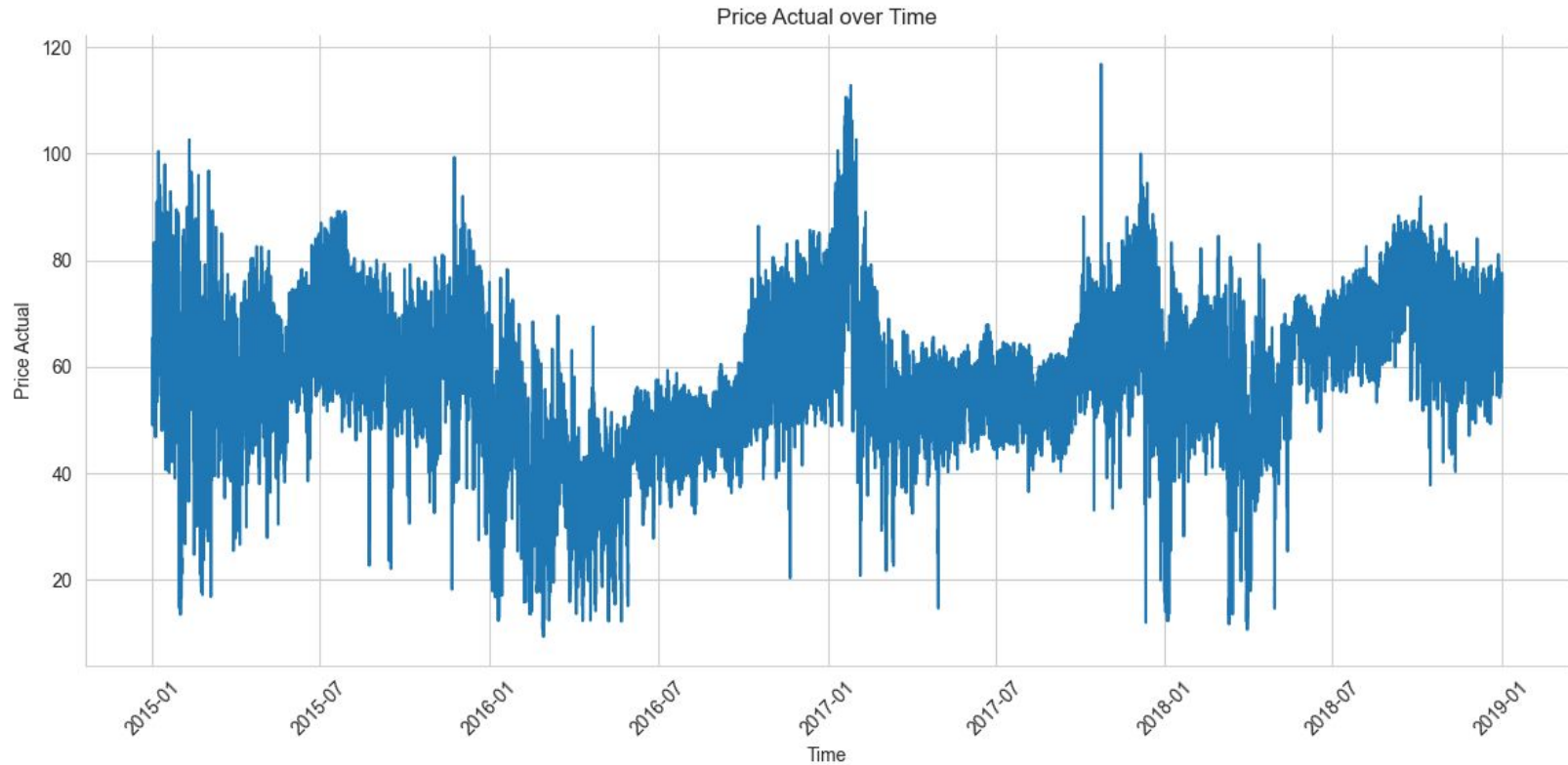
Develop a predictive model for forecasting electricity prices in Spain using historical energy and weather data.

Datasets

This dataset contains 4 years of electricity prices, energy generation and consumption, and weather data for Spain.

- Price: Obtained from Red Eléctrica de España (REE)
- Energy: Consumption and generation of various energy sources retrieved from European Network of Transmission System Operators for Electricity (ENTSOE).
- Weather: Data was obtained from the Open Weather API for the 5 largest cities in Spain.

Price Actual over Time



Price Actual Distribution Per Year



Price Actual Distribution Per Month



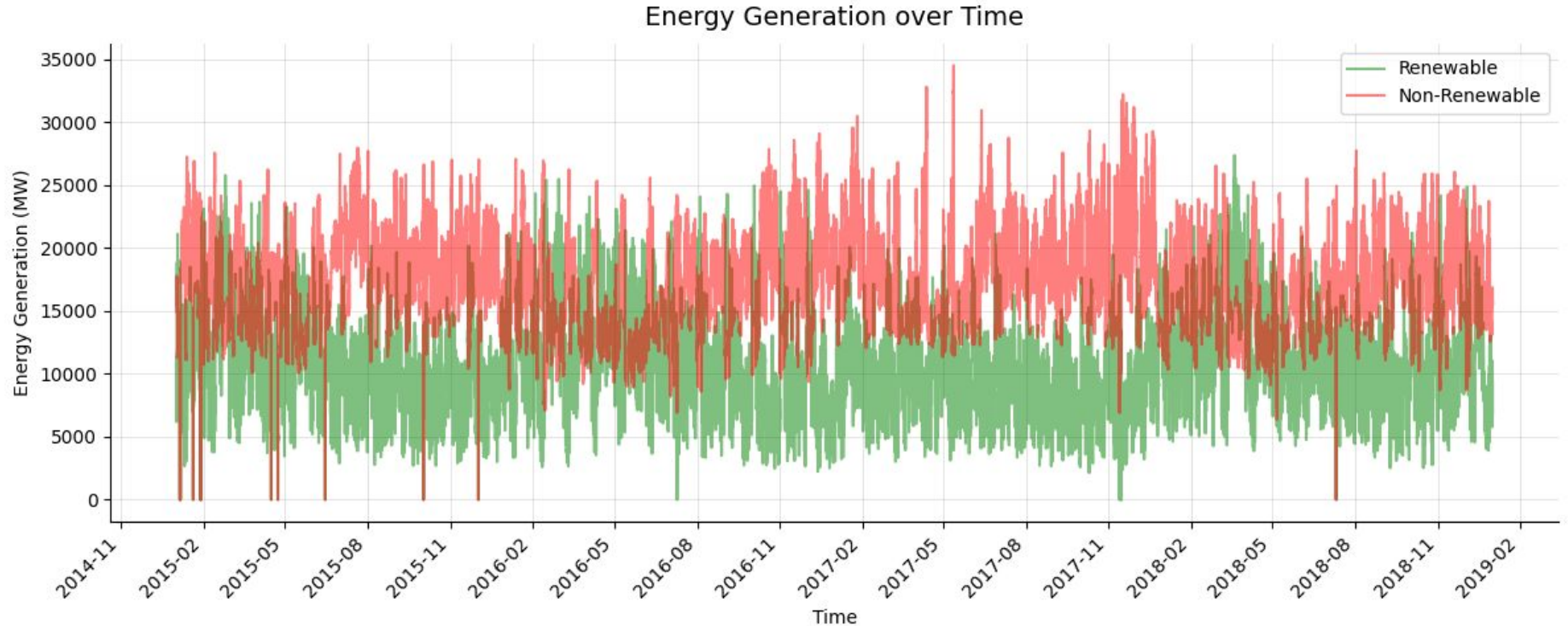
Energy Data



- 29 variables (including 2 related to price)
- 35,064 observations
- Data summary in the next slide

Variables	Description	Type
Time	Year:month:day:time in UTC	Continuous
Generation: Biomass / “Fossil Brown Coal/Lignite” / Fossil Gas / Hard Coal / Fossil Oil	Megawatt	Continuous
Generation Hydro: Pumped Storage Aggregated / Run-of-River and Poundage / Water Reservoir	Megawatt	Continuous
Generation: Nuclear/ Other / Other Renewable/ Solar / Waste/ Wind Onshore	Megawatt	Continuous
*Generation: Hydro Pumped Storage Aggregated / Marine / Wind Offshore/ Wind Offshore Day Ahead	Megawatt (dropped - whole column of data was missing)	Continuous
**Generation: Fossil Coal-derived Gas / Fossil Oil Shale / Fossil Peat / Geothermal	Megawatt (dropped - whole column of 0's)	Continuous
Forecast: Solar Day Ahead / Wind Onshore Day Ahead	Megawatt	Continuous
*Forecast Wind Offshore Day Ahead	Megawatt (dropped - whole column of data was missing)	Continuous
Total Load Forecast / Total Load Actual	Megawatt	Continuous
Price Day Ahead / Price Actual	Euro per Megawatt Hours	Continuous

Energy Data - A fight between renewable and non



Weather Data



- 17 variables
- 178,396 observations
- Data summary in the next slide

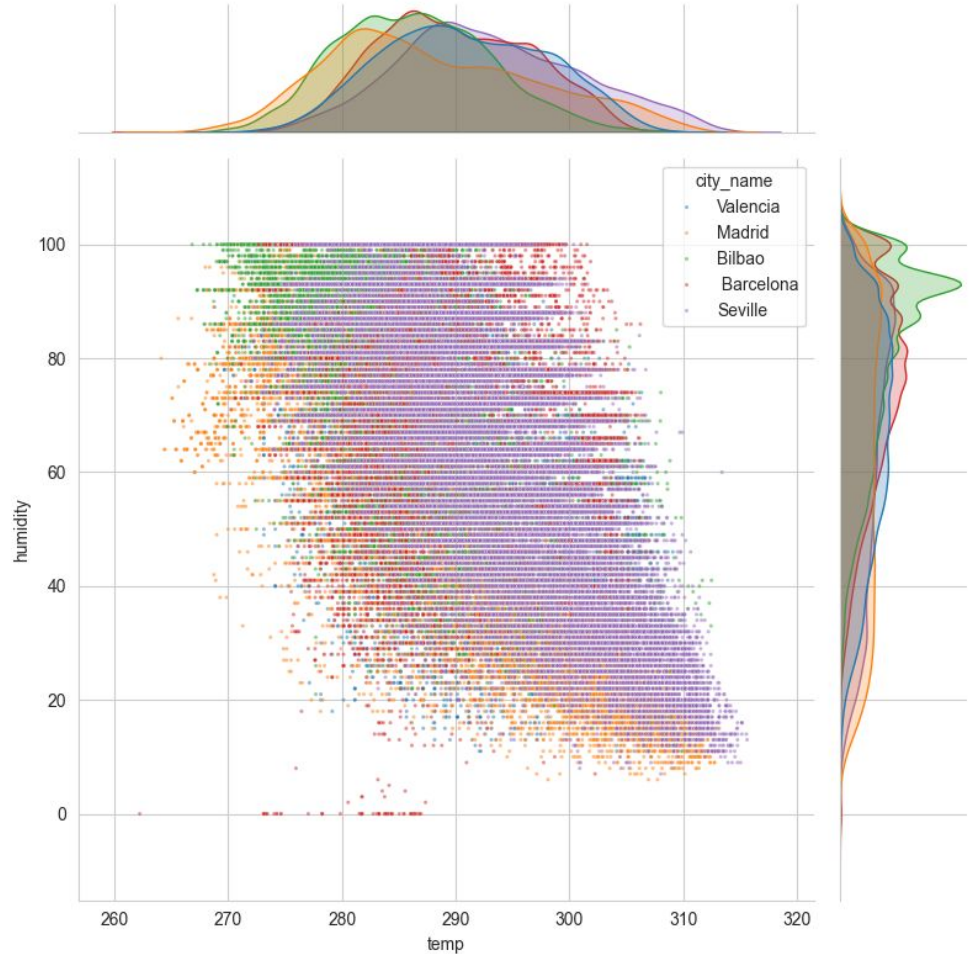
Variables	Description	Type
Time	Year month day time in UTC	Continuous
City Name*	Barcelona /Bilbao/Madrid/ Seville/Valencia	Nominal
Temperature/ Temp Min/ Temp Max	Kelvin	Continuous
Pressure	Hectopascal (hPa)	Continuous
Humidity	Percent	Continuous
Wind Speed	Meter per second	Continuous
Wind Degree	Wind Direction	Continuous
Rain 1h/ Rain 3h/ Snow 3h	Rain/Rain/Snow in the last 1/3/3 hour in millimeters	Continuous
Clouds All	Percent of cloudiness	Continuous
Weather Id*	Code used to describe weather	Discrete
Weather Main**	Rain/Snow/Clouds/Etc.	Nominal
Weather Description*	Long description	Nominal
Weather Icon*	Weather icon id for the website	Nominal

**the column was dropped after processing*

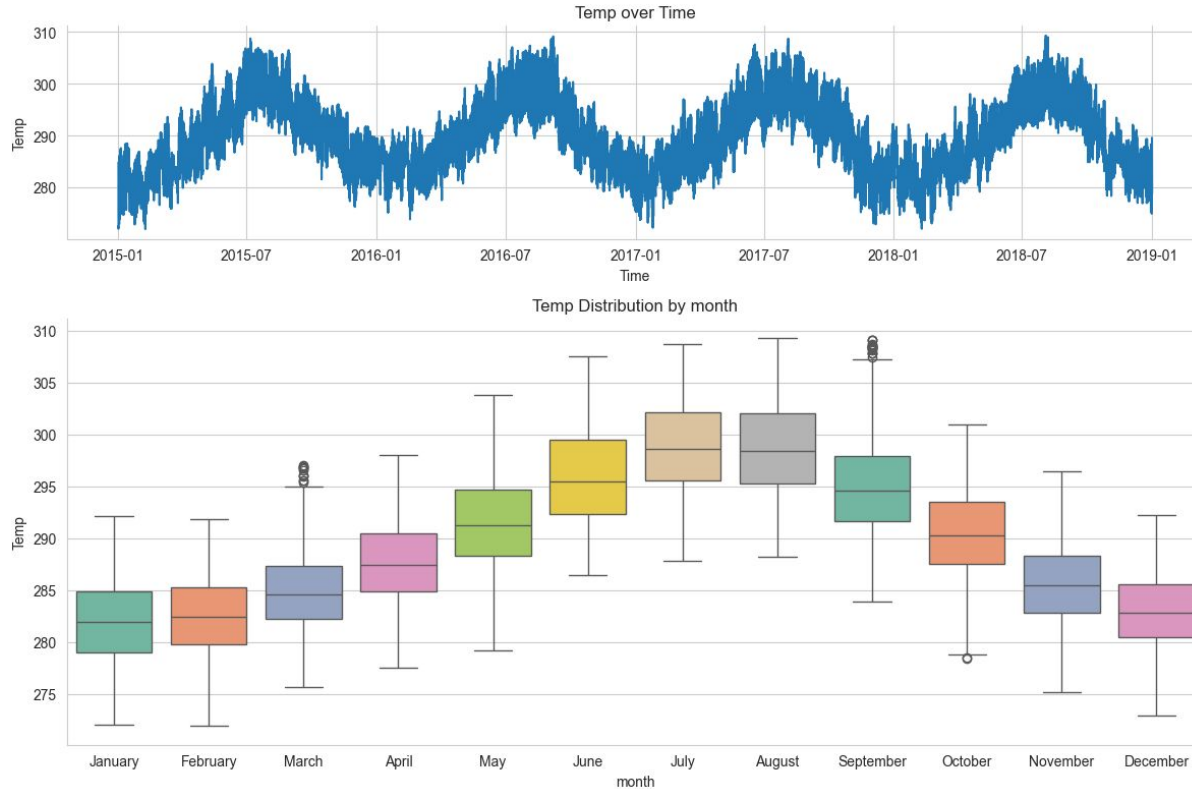
*** column was split into 5 columns after one-hot encoding*

Weather data - A tale of five cities

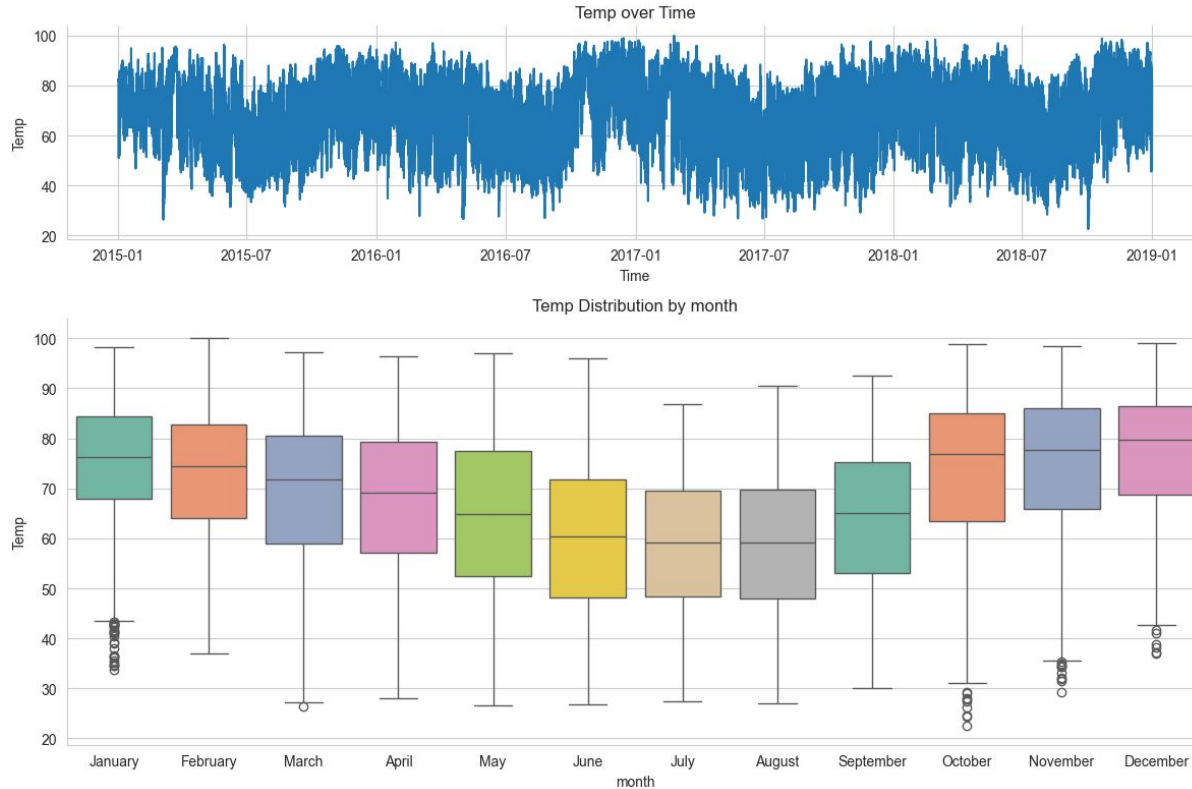
For each city, weather
observations are provided
per timestamp



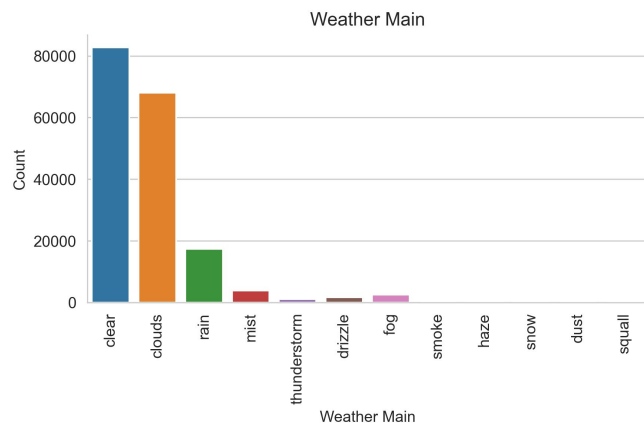
Weather Data - Temperature Distribution Per Month



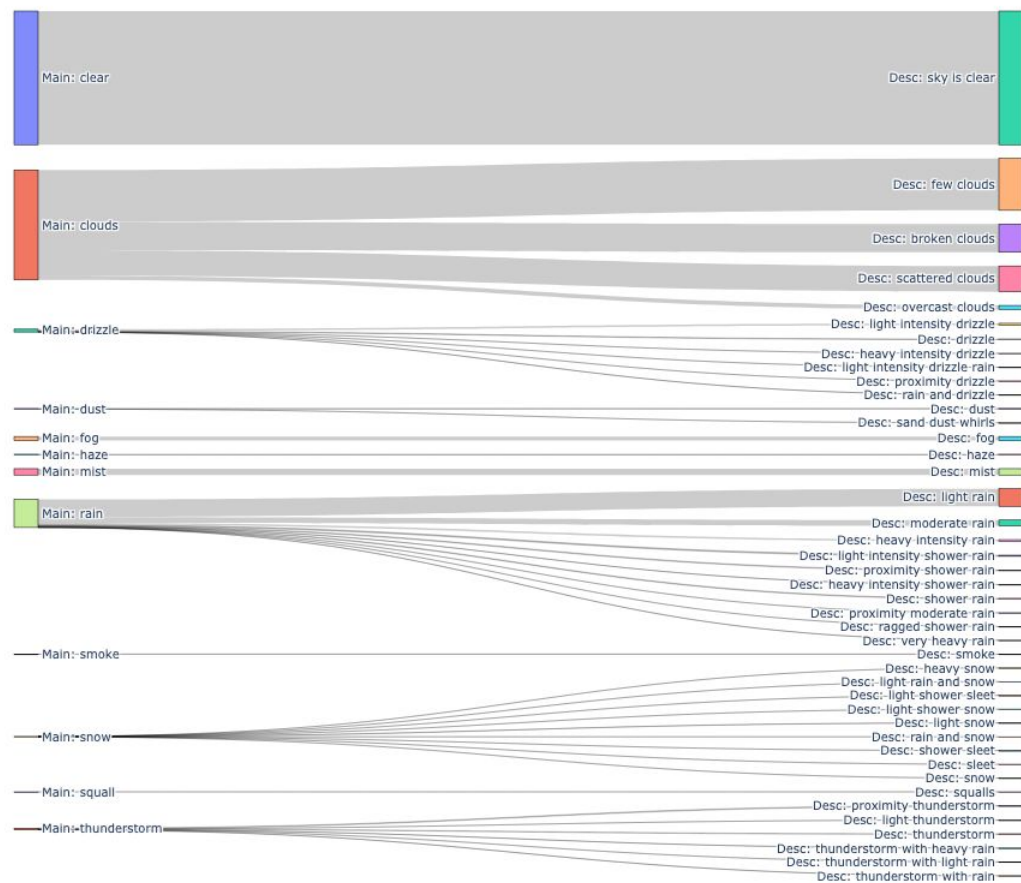
Weather Data - Humidity Distribution Per Month



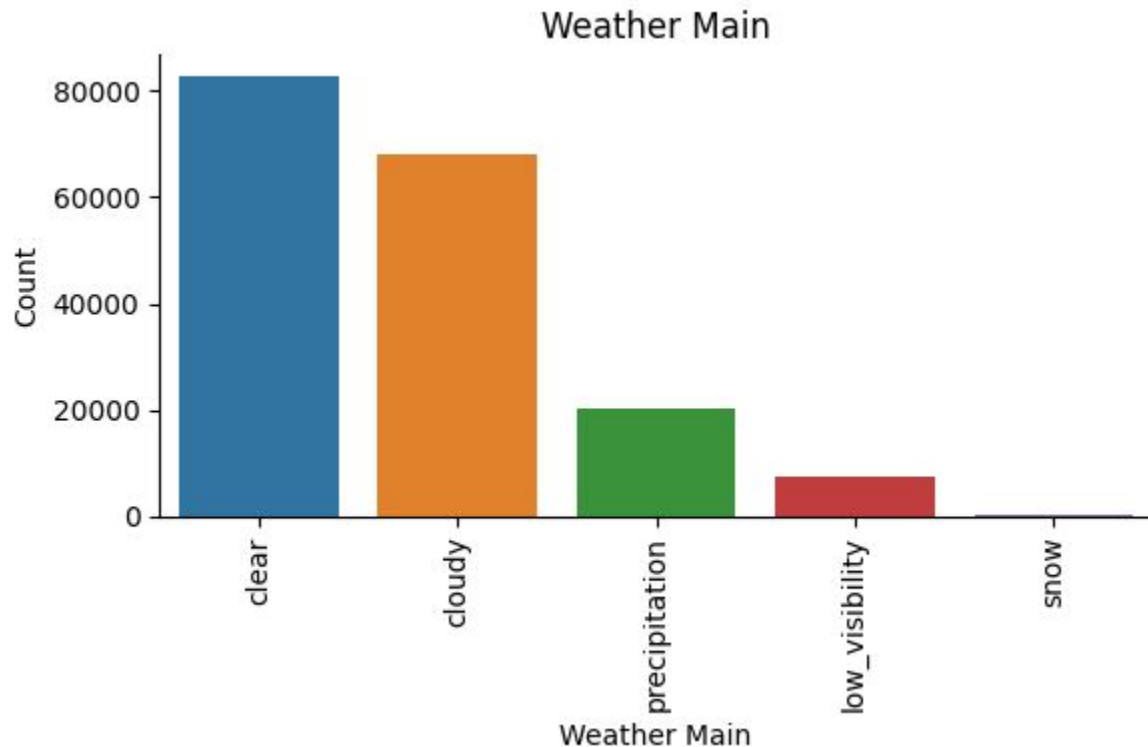
Weather Data - Distribution of Weather Main and Weather Description



Weather Main → Weather Description (No Loops)



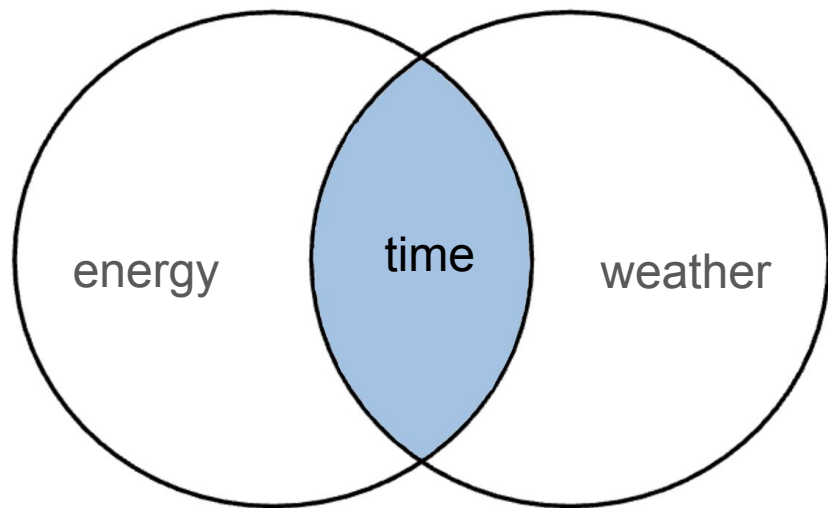
Weather Main Data After Encoding



Methods - Preprocessing

- Energy and Weather Data
 - Clean data (columns of all missing or zero)
 - Impute using moving average
- Weather
 - Normalize nominal weather data
 - Exam city-level information; average across cities
 - Aggregate by timestamp (one row per unique datetime)
 - One-hot encode nominal features
- Merge energy and weather tables
 - Inner join on time

Merged Dataset

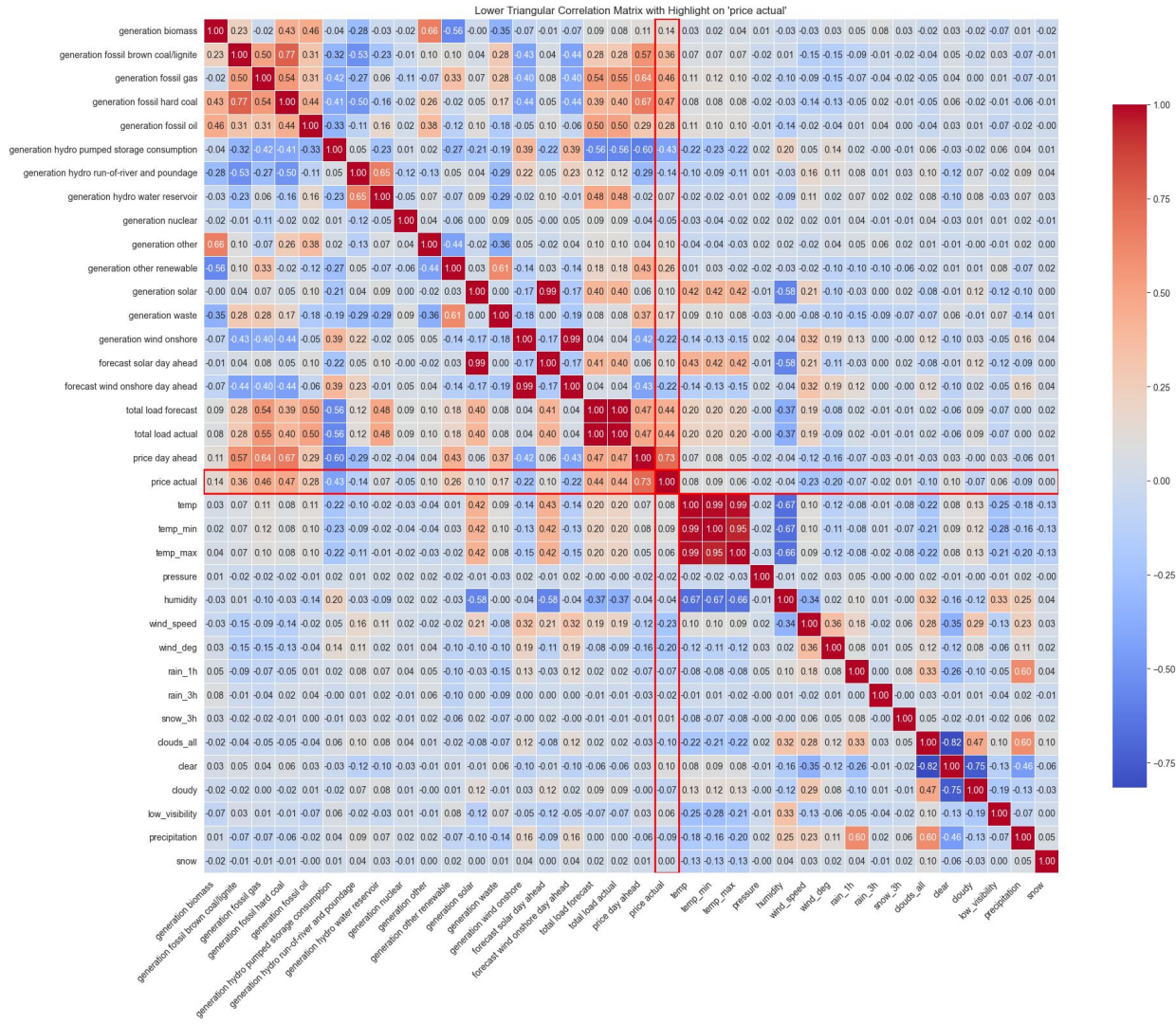


inner join

- 37 variables (including time)
 - 20 from energy
 - 16 from weather
- 35,064 observations
- Data summary in the next slide

Variables	Description	Type
Time	Year:month:day:time in UTC	Continuous
Generation : Biomass / "Fossil Brown Coal/Lignite" / Fossil Gas / Hard Coal / Fossil Oil	Megawatt	Continuous
Generation Hydro : Pumped Storage Aggregated / Run-of-River and Poundage / Water Reservoir	Megawatt	Continuous
Generation : Nuclear/ Other / Other Renewable/ Solar / Waste/ Wind Onshore	Megawatt	Continuous
Forecast : Solar Day Ahead / Wind Onshore Day Ahead	Megawatt	Continuous
Total Load Forecast / Total Load Actual	Megawatt	Continuous
Price Day Ahead / Price Actual	Euro per Megawatt Hours	Continuous
Temperature/ Temp Min/ Temp Max	Kelvin	Continuous
Pressure	Hectopascal (hPa)	Continuous
Humidity	Percent	Continuous
Wind Speed	Meter per second	Continuous
Wind Degree	Wind Direction	Continuous
Rain 1h/ Rain 3h/ Snow 3h	Rain/Rain/Snow in the last 1/3/3 hour in millimeters	Continuous
Clouds All	Percent of cloudiness	Continuous
Clear/ Cloudy/ Low visibility/ Precipitation/ Snow	Main weather condition	Discrete

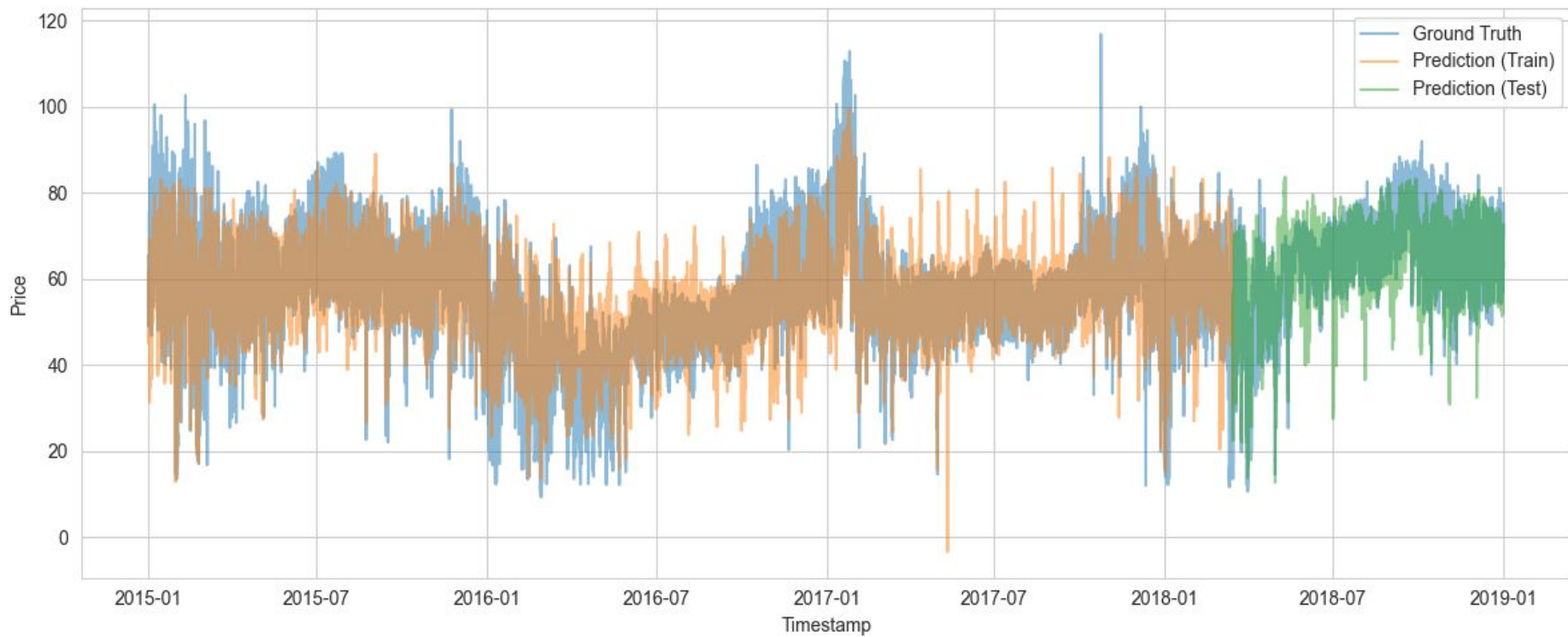
Merged Dataset - Lower Triangular Correlation with Highlight on “price actual”



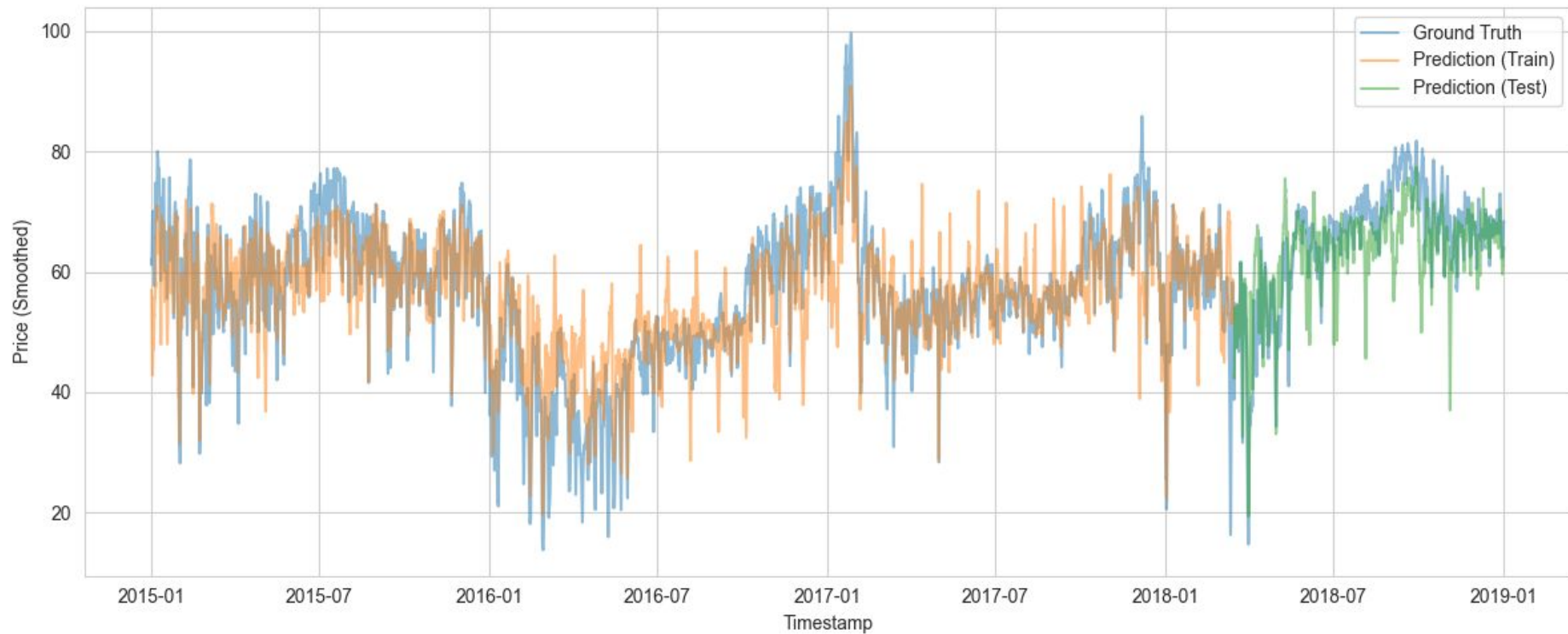
Methods - Modeling

- Train and Test split
 - 80% training, 20% test set
 - Cross-fold validations is employed for model validation
- Model Training
 - Linear regression
 - Ridge regression
 - Lasso regression
 - Support vector regression
 - Random forest regression
- Model Selection
 - Best performing model in term of RMSE is selected

Results



Results



Omnibus:	1918.247	Durbin-Watson:	0.126
Prob(Omnibus):	0.000	Jarque-Bera (JB):	9522.961
Skew:	-0.088	Prob(JB):	0.00
Kurtosis:	5.850	Cond. No.	2.47e+15

Results - Top 10 Variables with Strongest Impact on Price

Predictor	Coefficient	t-value
Price day ahead	0.7828	98.302
Wind speed	-2.0481	-38.802
Generation waste	-0.0242	-14.555
Temperature max	-1.1836	-10.565
Humidity	0.0723	10.548
Temperature	2.3515	9.991
Wind degrees	-0.0085	-8.197
Temperature min	-1.1225	-8.708
Snow in the last 3 hrs	3.4924	7.224
Low visibility	6.8720	6.664

Discussion

- Evaluate results:
 - MSE of 63.57
 - RMSE time-dependent patterns to improve accuracy
- What does it mean?
 - A significant relationship exists between energy demand, weather conditions, and electricity pricing
 - In particular, variables like total Load Forecast and Temperature strongly influence price behavior
- Future recommendations:
 - Advanced Modeling Techniques:
 - Explore more sophisticated methods including timeseries analysis to potentially reduce prediction errors.
 - Our current modeling only looks at each timestamp alone, we can include temporal information in a window on time to better model and understand changes in different attributes and their effect on price
 - City-Specific Modeling
 - Develop models using city-specific data to capture local variations and further refine the accuracy of the forecasts.

Resources

- Dataset:
 - https://www.kaggle.com/datasets/nicholasjhana/energy-consumption-generation-prices-and-weather/data?select=energy_dataset.csv
- Pictures:
 - <https://www.gasworld.com/story/envision-energy-unveils-1bn-spain-green-hydrogen-project/2144021.article/>
 - <https://www.travel-weather.net/spain/spain-weather-map.html>

Thank you for listening to
our presentation!